

HOW CAN A FUNCTION BE REPRESENTED

- **On a grid?** Favoured by scientist working with Numerical methods
 - + Easy and intuitive. Banded matrices. Finite expansion.
 - - Expensive. Not very accurate: numerical derivation and integration. Limited flexibility (a different grid gives a different equation).
- **Global analytical function?** As Harmonic Oscillator functions to describe something approximately an Harmonic oscillator. Or Gaussians. Traditional method used in Physics.
 - + Well chosen! → minimal expansion! Analytical integration and derivation. Accurate and potentially fast.
 - - Infinite expansion - where to stop? Usually not a banded matrix.

$$\Psi(x) = \sum_{i=1}^{\infty} c_i \phi_i(x)$$

HOW CAN A FUNCTION BE REPRESENTED

- **With localized analytical functions?** Examples: Finite Elements, **B-splines**.
- + Accuracy: Analytical integration and derivation. Adaptable to changing conditions. Finite expansion. Banded matrices.
- - Not as intuitive as e.g. a grid.

$$\Psi(x) = \sum_{i=1}^{\infty} c_i B_{i,k}(x)$$

B-SPLINES FOR EIGENVALUE PROBLEMS

$$H\Psi(x) = E\Psi(x).$$

With the B-spline expansion above it can be written:

$$H \sum_i c_i B_{i,k}(x) = E \sum_i c_i B_{i,k}(x).$$

Multiplying with $B_{j,k}(x)$ from the left and integrating we get

$$\sum_i c_i \int B_{j,k}(x) H B_{i,k}(x) dx = E \sum_i c_i \int B_{j,k}(x) B_{i,k}(x) dx.$$

Note

$$\int B_{j,k}(x) B_{i,k}(x) dx = 0 \text{ when } |i - j| \geq k$$

$$\mathbf{Hc} = E\mathbf{Bc}$$

a generalized eigenvalue problem

THE HYDROGEN ATOM

FROM THREE DIMENSIONS TO ONE

$$H = \frac{\mathbf{p}^2}{2m_e} - \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r} = -\frac{\hbar^2 \nabla^2}{2m_e} - \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r}.$$

Central symmetric potential! Separation of variables.

$$\Psi(r, \theta, \phi) = R_{nl}(r) Y_{\ell m}(\theta, \phi) = \frac{P_{nl}(r)}{r} Y_{\ell m}(\theta, \phi).$$

- Spherical Harmonics: $Y_{\ell m}(\theta, \phi)$

$$\nabla^2 \left(\frac{P_{nl}(r)}{r} Y_{\ell m}(\theta, \phi) \right) = \frac{1}{r} \left(\frac{d^2}{dr^2} - \frac{\ell(\ell+1)}{r^2} \right) P_{nl}(r) Y_{\ell m}(\theta, \phi)$$

$$\left(-\frac{\hbar^2}{2m_e} \frac{d^2}{dr^2} + \frac{\hbar^2 \ell(\ell+1)}{2m_e r^2} - \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r} \right) P_{nl}(r) = E P_{nl}(r).$$

BOUNDARY CONDITIONS

$$R_{n\ell}(r) = \frac{P_{n\ell}(r)}{r} \rightarrow P_{n\ell}(0) = 0$$

bound states $\rightarrow P_{n\ell}(r) \rightarrow 0$ when $r \rightarrow \infty$

with a big enough (spherical) box we can assume that the function is zero at both boundaries!

- big enough box
- remove the first and last B-spline

PRACTICAL UNITS

For calculations on atomic systems it is practical to write programs in Hartree units. The base units are then:

$$\begin{aligned}\text{lenght : } a_0 &= \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} \approx 0.5 \cdot 10^{-10} \text{m} \\ \text{mass} &= m_e \approx 9.1 \cdot 10^{-31} \text{kg} \\ \text{energy : 1Hartree} &= \frac{m_e e^4}{(4\pi\epsilon_0)^2 \hbar^2} = \frac{\hbar^2}{a_0^2 m_e} \approx 27.2114 \text{eV} \\ \text{time} &= \frac{(4\pi\epsilon_0)^2 \hbar^3}{m_e e^4} = \frac{a_0^2 m_e}{\hbar} \approx 2.42 \cdot 10^{-17} \text{s}.\end{aligned}$$

THE KNOT SEQUENCE

- try different k and different number of knots
- first knot in $r = 0$ - with k knots
- $R_{max} = 10$ Bohr radii ? or do you want to have it larger?
- linear? or exponential? or a combination?
- explore the freedom to use different knot sequences!

Note that

$$\int B_{j,k}(r) \left(-\frac{d^2}{dr^2} \right) B_{i,k}(r) dr = \int \frac{dB_{j,k}(r)}{dr} \frac{dB_{i,k}(r)}{dr} dr,$$

The derivatives of the splines are exactly known -no extra errors introduced!

HOW TO INTEGRATE?

- Grid based method $\rightarrow$ trapezoid rule $\sum_i f(x_i)h$
- We know the function everywhere (not only at the grid points)
use *Gaussian Quadrature to integrate with machine accuracy!*

GAUSSIAN QUADRATURE

Approximation of the integral as a **weighted sum** of function values at **specific points** (*abscissas*). An n -point Gaussian quadrature rule is constructed to give an *exact* result for polynomials of degree $2n - 1$ or less by a suitable choice of the **abscissas** x_i and **weights** w_i for $i = 1, \dots, n$;

$$\int_a^b W(x)f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \text{ compare } \sum_{i=1}^{all} f(x_1 + h(i-1))h$$

Exact if $f(x)$ is a polynomial. Different $W(x) \rightarrow$ different weights and abscissas. Named after the class of orthogonal polynomials whose roots define the abscissas. $W(x) = 1$ are integrated with **Gauss-Legendre** quadrature (sometimes just called Gauss). Since the B-splines are polynomials, that is what we will use here. You find several other possibilities in *Numerical Recipes*

WHAT TO DO!

- Solve for a few different ℓ -values. Compare with the known values. How many bound states of hydrogen can you find? How is this affected by the choice of knot-sequence? R_{max} ?
- Check also the eigenfunctions. What about normalization?
- If you have time: point charge $\rightarrow$ spherical charge distribution

$$\begin{aligned} V(r) &= \frac{-Ze^2}{4\pi\epsilon_0} \frac{1}{2R_0} \left(3 - \left(\frac{r}{R_0} \right)^2 \right), \quad r < R_0 \\ &= \frac{-Ze^2}{4\pi\epsilon_0} \frac{1}{r}, \quad r > R_0. \end{aligned}$$

- 1) Highly charged ions and the real nucleus?
- 2) Hydrogen: If the nucleus was not a point but a large sphere ($R_0 \approx 1.5$ Bohr radii). What would change?

THINGS TO TEST

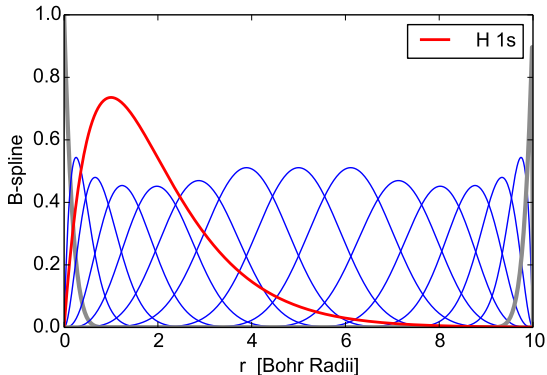


FIGURE : The B-splines are generated on a linear knot sequence. The first and last Bsplines are omitted. The remaining B-splines are used to express the functions: for example the hydrogen ground state. Can it be well represented by a sum over the blue curves?

THINGS TO TEST

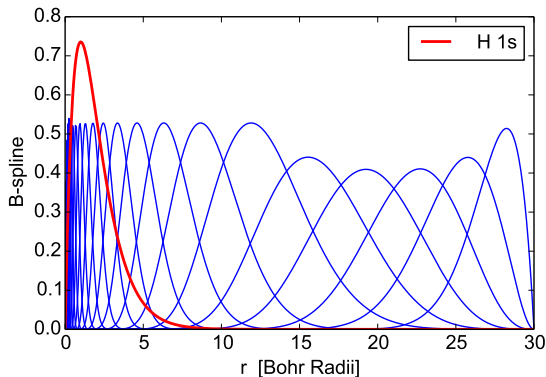


FIGURE : The B-splines are generated on an exponential knot sequence. The **B-splines** are used to express the functions: for example **the hydrogen ground state**. Can it be well represented by a sum over the blue curves?